Oxidation & Reduction

Exam code: 0620 & 0971

Download PDF

Written by: Alexandra Brennan
Reviewed by: Stewart Hird
Updated on 30 September 2025

Did this video help you? Yes No

Oxidation & reduction

Roman numerals and oxidation numbers

- Transition elements can bond in different ways by forming ions with different charges
- When naming, the charge on the ion is shown by using a Roman numeral after the element's name
- For example, iron can form different ions
  - Iron ions with a 2+ charge, Fe^{2+}, are called **iron(II)** ions
  - Iron ions with a 3+ charge, Fe^{3+}, are called **iron(III)** ions

- The Roman numeral is the **oxidation number** of the element
- For example, iron reacts with oxygen to form iron oxide
  - But the name and formula of the product depend on the oxidation state of the iron ions
  - Iron(II) ions
    - The iron oxide product contains iron(II) ions, Fe^{2+}
    - The name of this product is iron(II) oxide
    - The formula of this product is FeO
  - Iron(III) ions
    - The iron oxide product contains iron(III) ions, Fe^{3+}
    - The name of this product is iron(III) oxide
    - The formula of this product is Fe_2O_3

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

1. State the oxidation number of the transition metal ion in silver(I) chloride, AgCl.
2. Name, including Roman numeral, the compound with the formula $CoCl_2$.

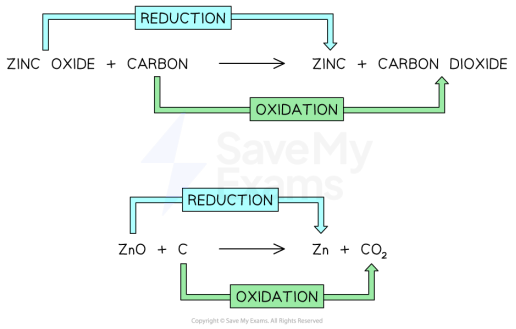
Answers:

1. The oxidation number of the transition metal ion in silver(I) chloride, AgCl, is 1
   - **Remember:** The Roman numeral gives the oxidation number of the element before
2. The name, including Roman numeral, of the chemical with the formula $CoCl_2$ is cobalt(II) chloride
   - $CoCl_2$ contains two chloride ions, each with a 1– charge
   - So, the cobalt ion has a 2+ charge
   - This means the compound contains cobalt(II) ions, Co^{2+}
   - Therefore, it is called cobalt(II) chloride

What is a redox reaction?

- A **redox** reaction is where oxidation and reduction take place together at the same time in the same reaction
- In terms of oxygen:
  - **Oxidation** is where **oxygen** is **added** to an element or a compound
  - **Reduction** is where **oxygen** is **removed** from an element or compound

- The reaction between zinc oxide and carbon is an example of a redox reaction
  - Zinc oxide is reduced because it has **lost** oxygen
  - Carbon is oxidised because it has **gained** oxygen



Both reduction and oxidation has occurred in this reaction so it is classed as a redox reaction

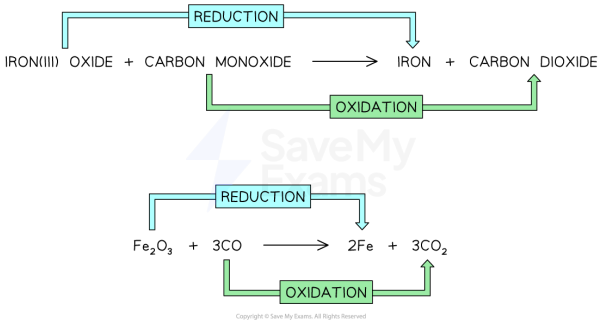
Worked Example

iron(III) oxide + carbon monoxide → iron + carbon dioxide

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

Answer:

- Iron(III) oxide loses oxygen, so it is reduced
- Carbon monoxide gains oxygen, so it is oxidised



Examiner Tips and Tricks

You may see the term oxidation state used instead of oxidation number.

Although there is a subtle difference between the two terms (this is beyond the scope of this course), they are often used interchangeably.

Usually oxidation number is used to refer to the Roman numerals found within the name.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Test yourself Flashcards

Was this revision note helpful? Yes No

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.